

UBT618: DOWNSTREAM PROCESSING

L	T	P	Cr
3	0	0	3.0

Course Objective: The course provides fundamental concepts of various downstream purification steps involved in a bioprocess industry.

Syllabus

Introduction: Basic concepts of separation technology, overview of major upstream and downstream processes, importance of downstream processing in biotechnology, economic evaluation of downstream processing, separation characteristics of biological molecules, generic scheme of bioseparation, modern separation technology in bioprocessing.

Primary Isolation and Recovery: Selection of purification methodologies, biomass removal and disruption by physical, chemical and biological methods, Types of Homogenizers, Types of filters (vacuum filter, plate and frame filter, leaf filter), Advanced Centrifugation, Theory of centrifugation, Types of centrifuges (tubular bowl centrifuge, basket centrifuge, ultracentrifuge), Precipitation, Coagulation and flocculation.

Membrane based Separation, Extraction and Adsorption: Membrane process, ultrafiltration, nanofiltration, reverse osmosis, dialysis, Extraction, liquid-liquid extraction, Batch extractions, staged extractions, solvent recovery, applications of extraction. Evaporation, Types of evaporation, Adsorption, adsorbents types, their preparation and properties, types of adsorption isotherms and their importance in bioprocessing, adsorption in fixed bed.

Chromatography, Drying and Case Studies: General theory, partition coefficient, types of chromatography: Ion exchange, gel permeation, affinity, HPLC, Crystallization: Batch and continuous crystallization, crystallization equipment, Principles of drying (Water content in biological products), Heat transfer modes: Conduction, Natural and Forced Convection, Radiation, Design and Operation of Drying Equipment: Hot-air oven, Spray dryer, Vacuum dryer, Freeze dryer, Case studies: Downstream processing of baker's yeast and citric acid.

Laboratory Work (if applicable)

Course Learning Objectives (CLO)

The students will be able to:

1. Comprehend the interplay between upstream and downstream processes.
2. Describe the main stages of downstream processing operations.

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3. Explain the principles of major downstream operations used in a bioprocess industry such as filtration, centrifugation, extraction and chromatography.
4. Apply different techniques such as precipitation, coagulation and flocculation in downstream processing
5. Design recovery outline in polishing of a product employing crystallization and drying methods.

Text Books

1. Belter PA, Cussler E and Hu WS, Bioseparation – Downstream Processing for Biotechnology, Wiley Interscience (1988).
2. Harrison RG, Bioseparations: Science and Engineering, Oxford University Press (2015).
3. Sivasankar B, Bioseparations: Principles and Techniques, PHI Learning Pvt. Ltd. (2006).

Reference Books

1. Ahuja S, Handbook of Bioseparations, Academic Press (2000).
2. G. Subramanian-Bioseparation and Bioprocessing: Biochromatography, membrane separations, modeling, validation, Vol-I, Wiley-VCH Verlag GmbH Germany (1998).

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1	MST	25-35
2	EST	35-45
3	Sessional (May include Assignments/Projects/Quizzes)	30

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